

SARASWAT ACADEMY
CLASS-IX- MATHEMATICS
CHAPTER-2: Introduction to
Linear Polynomials

MCQ'S

Note: Each question has 1 mark

- Q1: Which one of the following is constant polynomial?
(a) $4x^2 + 2$ (b) $7x^0 + x^3 + 22$
(c) $x^0 + x \cdot 0$ (d) $4x$
- Q2: The value of the quadratic polynomial $4a^2 - 2a + 3$ at $a = -1$
(a) -9 (b) 9 (c) 10 (d) -10
- Q3: An auto-rikshaw fare starts at Rs. 25 and remains the same for the initial 2 km. Then it increases by Rs. 15 per km. What will be the fare for a travel of 5 km?
(a) Rs. 50 (b) Rs. 60 (c) Rs. 70 (d) Rs. 80
- Q4: If we change the values of a, keeping b fixed in the equation $y = ax + b$, the slope of the line changes while the y-intercept remains
a) Fixed b) Changes with 'a' c) pass through origin d) None of these
- Q5: **Assertion (A):** In the equation $y = ax + b$, a represents the slope of the line and b represents the y-intercept.
Reason (R): If we change the values of b keeping the value of a fixed, the lines shift but remain parallel to the original line.

COMPETENCY BASED QUESTIONS

Note: Each question has 3 marks

- Q1: If you have Rs. 400 and you save Rs. 150 every month, find the total amount you have after (i) 6 months (ii) 2 years. Express this as a linear pattern.
- Q2: Ruby has 3 times as many two-rupee coins as she has five rupee-coins. If she has a total Rs. 88, how many coins does she have of each type?
- Q3: A telecom company charges Rs. 750 for a certain recharge scheme. This prepaid balance is reduced by Rs.25 each day after the recharge. After how many days will the balance run out?
- Q4: Draw the graphs of $y = 4x$, $y = 2x$ and $y = x$. In each case, reflect on the role of 'a' and 'b'
- Q5: A gym charges a fixed monthly fee and an additional cost per hour for using the badminton court. A student using the gym observed that when she used the badminton court for 10 hours, her bill was Rs. 800. When she used it for 15 hours, her bill was Rs. 1100. If the monthly bill y depends on the hours of the use

of the badminton court, x , according to the relation $y = ax + b$, find the values of a and b .

CASE STUDY

A telecom company charges a fixed monthly fee and an additional cost per GB of the internet data used. A student observes that when she used 10 GB, her bill was Rs. 350. When she used 20 GB, her bill was Rs. 550. If the monthly bill y depends on the amount of data used, x (in GB), according to the relation $y = ax + b$, find the values of a and b . Note that x = number of GB of the internet data used and y = total monthly cost in Rupees.

Based on the case study "Analyzing Monthly Telecom Expenses". Answer the following questions

1. If the student receives a bill for ₹750, how many GB of data did she consume that month? Show your work by setting up a linear equation.
2. If the telecom company decides to double the fixed monthly fee but keeps the data rate the same, how would the graph of the new bill compare to the original line?
3. Does this telecom bill model represent linear growth or linear decay? Justify your answer based on the sign of the slope.
4. Using the equation as a function, calculate the total bill () if the student uses exactly 30 GB of data.